IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

EEE/EIE PART III/IV: MEng, BEng and ACGI

DIGITAL SIGNAL PROCESSING

Corrected copy

Wednesday, 13 December 9:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : W. Dai

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DIGITAL SIGNAL PROCESSING

1. a) Figure 1.1 shows a digital signal processing system containing 4 subsystems, A, B, C and D with impulse responses as shown in Table 1.

Subsystem	Impulse response
A	$h_a(n)$
B	$h_b(n)$
C	$h_c(n)$
D	$h_d(n)$

Table 1

Write an expression for $h(n)$, where $h(n)$ is the impulse response of the complete LTI system with input $x(n)$ and output $y(n)$. [2]

Determine $h(n)$ for the following definitions of $h_a(n)$, $h_b(n)$, $h_c(n)$, $h_d(n)$. [3]

$$h_a(n) = \{0.5, 0.3, 0.5\}$$
$$h_b(n) = (n+1)u(n)$$
$$h_c(n) = h_b(n)$$
$$h_d(n) = \delta(n-1).$$

Determine the output of the LTI system, $y(n)$, when the input is

$$x(n) = \delta(n+2) + 3\delta(n-1) - 4\delta(n-3).$$

[4]

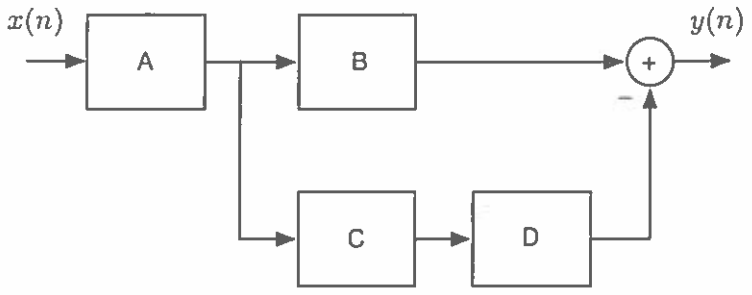


Figure 1.1

b) An LTI system, with input signal $x(n)$ and output signal $y(n)$ is characterized by the difference equation

$$y(n) = 0.7y(n-1) - 0.1y(n-2) + 2x(n) - x(n-2).$$

Determine the impulse response and the unit step response of this system. [6]

c) Consider two classes of discrete-time systems: recursive and nonrecursive. State the important differences between these two classes of systems. [1]

Derive the difference equation of a recursive system for which the output $y(n)$ is the cumulative average of a signal $x(n)$. Draw the signal flow diagram of this system. [4]

2. a) i) Given an M-fold decimator with input $x(n)$ and output $y_D(n)$, write down the expression for $y_D(n)$ in terms of $x(n)$. [2]

ii) Given an L-fold expander with input $x(n)$ and output $y_E(n)$, write down the expression for $y_E(n)$ in terms of $x(n)$. [2]

b) Next consider the multirate signal processing system shown in Fig. 2.1.

i) Write down $Y(z)$ in terms of $X(z)$ for this system. [4]

ii) In Fig. 2.2, the magnitude spectrum of $X(z)$ is shown for the frequency range $-\pi \leq \omega \leq \pi$. Now consider the case when the LTI system $H(z)$ is an ideal lowpass filter with normalized cut-off frequency $\frac{\pi}{2}$.

Draw a labelled sketch of the magnitude spectrum of $Y(z)$ covering the range of frequencies $-2\pi \leq \omega \leq 4\pi$. Explain the main features of this magnitude spectrum sketch.

Does aliasing occur? Explain your answer. [4]

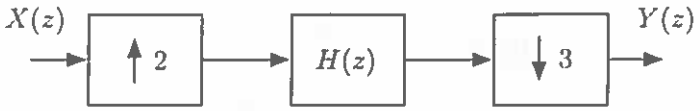


Figure 2.1

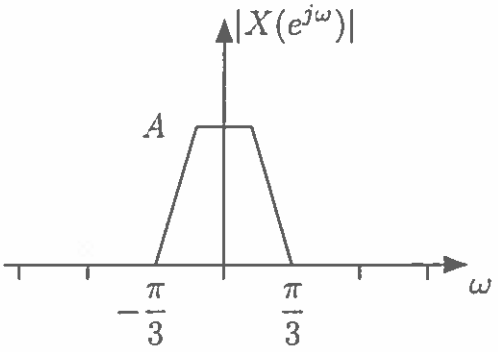


Figure 2.2

c) State the Noble identities relating to multirate signal processing. Hence, redraw the system in Fig. 2.1 such that the computation of $H(z)$ is performed at the lowest possible sampling frequency. [4]

d) i) Write down an expression for the 3-phase Type 1 polyphase decomposition of a filter $H(z)$.

ii) Redraw Fig. 2.1 using the 3-phase Type 1 polyphase decomposition.

iii) Redraw your solution to part c) using 3-phase Type 1 polyphase filtering.

iv) State the relative merits of the structures resulting from (ii) and (iii).

[4]

3. a) Using a few sentences and one or more relevant diagrams, describe the similarities and differences between the continuous-time Fourier transform, the DTFT and the DFT? [6]

b) i) With discrete-time index n, consider a signal $x(n)$ for $n = 0, 1, \ldots, 5$. State and explain a sufficient condition on $x(n)$ such that its Fourier transform $X(e^{j\omega})$ is real valued. [2]

ii) Find $x(n)$ for $X(e^{j\omega}) = \{6, -5, 3, 4, 3, -5\}$. [5]

c) A continuous-time analogue signal containing an audio recording of bird song has a bandwidth of 4 kHz. It is required to compute the spectrum of this signal using an N-point DFT with $N = 2^m$, for integer m, and a frequency resolution $\Delta f \leq 50$ Hz.

i) What is the minimum sampling frequency required?

ii) What is the minimum number of samples required?

iii) What is the minimum duration of the analogue signal required?

[3]

d) By exploiting the DFT, prove that

$$\sum_{l=-\infty}^{\infty} \delta(n + lN) = \frac{1}{N} \sum_{k=0}^{N-1} e^{j2\pi kn/N}.$$

[4]

4. a) Consider a general discrete-time system with system function

$$H(z) = \frac{Y(z)}{X(z)}.$$

With reference to the polynomials $X(z)$ and $Y(z)$, explain the meaning of, and define criteria for, the underlined terms in the following statements.

i) The system $H(z)$ is an unstable system.

ii) The system $H(z)$ is a non-causal system.

iii) The system $H(z)$ is a maximum phase system.

[3]

b) Consider a linear time-invariant system

$$H(z) = \frac{(z-\frac{1}{3})(z+2)(z^2+\frac{1}{9})}{(z^2+2z+5)(z^2-4z+13)}.$$

i) Find the roots of $H(z)$ and draw a labelled sketch showing the roots on the z-plane. [6]

ii) Determine any regions of convergence of $H(z)$ and state whether the inverse z-transform associated with each region of convergence is left-sided, right-sided or two-sided. Comment on the stability of $H(z)$. [5]

c) Given the signal representation in the z-domain

$$X(z) = z^3 + 2z^2 - 2 + \frac{2}{z^2+5z+4}, \quad |z| > 4$$

find the corresponding signal representation in the time domain using the inverse z-transform. [6]